

//reader writers problem

```
#include<semaphore.h>
#include<stdio.h>
#include<stdlib.h>
#include<unistd.h>
#include<pthread.h>
sem_t mutex, wrt;
int sharedvar = 99;
pthread_t writers[5], readers[5];
int readercount = 0;
void reader()
{
    sem_wait(&mutex);
    readercount++;
    if(readercount == 1){
        sem_wait(&wrt);
        sem_post(&mutex);
    }
    printf("\n%d reader is reading in sharedvar = %d", readercount, sharedvar);
    sem_wait(&mutex);
    readercount--;
    if(readercount == 0){
        sem_post(&wrt);
        sem_post(&mutex);
    }
}
void writer()
{
    sem_wait(&wrt);
    printf("\nwriter has entered Critical section");
    sharedvar++;
    printf("\nWriter changed the value of sharedvar to %d", sharedvar);
    sem_post(&wrt);
}
void main()
{
    int n2, i;
    printf("\nEnter the number of readers and writers");
    scanf("%d", &n2);
    sem_init(&mutex, 0, 1);
    sem_init(&wrt, 0, 1);
    for(i = 0; i < n2; i++)
    {
        pthread_create(&writers[i], NULL, (void*)writer, NULL);
        pthread_create(&readers[i], NULL, (void*)reader, NULL);
    }
    for(i=0; i < n2; i++){
        pthread_join(writers[i], NULL);
        pthread_join(readers[i], NULL);
    }
}
```

Output:

Enter the number of readers and writers2 3

writer has entered Critical section

Writer changed the value of sharedvar to 100

1 reader is reading in sharedvar = 100

writer has entered Critical section

Writer changed the value of sharedvar to 101

1 reader is reading in sharedvar = 101

```
Enter the number of readers and writers2 3
```

```
writer has entered Critical section
```

```
Writer changed the value of sharedvar to 100
```

```
1 reader is reading in sharedvar = 100
```

```
writer has entered Critical section
```

```
Writer changed the value of sharedvar to 101
```

```
1 reader is reading in sharedvar = 101%
```